Regular black holes with cold remnants at a 100 TeV collider: greybody emission, a full uncertainty budget and the discovery reach

Project Baby Universe — Genesis Branch
(draft for external review; not submitted)

31 October 2026

Abstract

If gravity becomes strong at a few TeV in $D = 4 + n$ dimensions and its short-distance completion is a quasi-topological curvature tower, collider black holes are regular, evaporate towards a zero-temperature extremal remnant, and appear as a hard jet recoiling against an invisible remnant. We simulate production with real parton distributions, Hawking emission with greybody factors computed on the actual regular metric, Pythia showering, an FCC-hh detector model with pile-up $\mu = 1000$, and a profile-likelihood analysis validated against the ATLAS 139 fb^{-1} monojet background ($K = 1.18$, 10% rms shape residual). For $n = 6$ and $M_D = 8$ TeV the 5σ discovery reach at 100 TeV with 30 ab^{-1} is $M_{\rm ext} = 33.4^{+0.8}_{-1.1}$ TeV and the expected 95% CL exclusion is $35.0^{+1.2}_{-1.0}$ TeV, with a documented budget of scale, PDF, normalisation, electroweak, emission-model and detector variations. Near extremality the regular holes emit gauge bosons at only 7% of the blackbody rate. The signal is separated from a GR black hole with the same threshold by 13 selected events, a result that does not depend on the emission model. These are exterior observables; they test the regular-black-hole tower, not any statement about the interior.

1 Model and generator

Holes form when the trapped mass $M = y_n\sqrt{\hat{s}}$ (head-on area bound) exceeds the extremal mass $M_{\rm ext} = a_n M_D \mu_{\rm ext}(c_l)$ of the tower with length scale $l = c_l/M_D$. The production cross-section uses the NNPDF2.3 LO parton luminosity times the geometric cross-section with the Yoshino–Nambu form factor. Emission is Monte Carlo Hawking radiation of Standard Model species on the brane until the remnant is reached; the temperature vanishes at $M_{\rm ext}$.

Greybody factors. The brane metric is $-f\,dt^2 + dr^2/f + r^2 d\Omega_2$ with the model-U $f(r) = 1 - r^2\psi(\mu/r^{D-1})$. We solve the spin-0, $\frac{1}{2}$ and 1 master equations ($V = f[l(l+1)/r^2 + f'/r]$, $W^2 + W'$ with $W = k\sqrt{f}/r$, and $f\,l(l+1)/r^2$) with ingoing conditions at r_+. Checks: $\Gamma_0 \to 4r_+^2\omega^2$ (horizon area), $\sum(2j+1)\Gamma_j \to (\omega b_c)^2$ (geometric optics), and agreement with an independent Schwarzschild code to 10^{-4}.

Proposition 1 (numerical)**.** *For the $D = 10$ Hayward tower, the per-degree-of-freedom number flux relative to the geometric blackbody is (spin 0, $\frac{1}{2}$, 1) = $(1.42, 1.01, 0.07)$ at $\mu = 1.05\mu_{\rm ext}$, $(1.29, 1.42, 0.94)$ at $3\mu_{\rm ext}$ and $(1.26, 1.32, 1.09)$ at $10\mu_{\rm ext}$ (Tangherlini $n = 6$: $1.23, 1.19, 1.22$). With Standard Model degrees of freedom the luminosity is 0.92–1.32 of the geometric value and the spin-1 share of quanta falls from 26% to 2% near the remnant.*

2 Detector, backgrounds and statistics

- *Backgrounds*: $Z(\nu\nu)$+jet, $W(\ell\nu)$+jet, $t\bar{t}$ and QCD multijets in p_T slices (Pythia 8.317, NNPDF2.3 QCD+QED LO), with a normalisation $K = 1.18$ fitted at 13 TeV to the ATLAS

monojet background.

- *Detector*: Delphes FCC-hh card parameters (resolutions, acceptance to $|\eta| < 6$), a jet-response tail, pile-up $\mu = 1000$ with charged-hadron subtraction and a neutral residual.
- *Selection*: MET > 1 TeV, leading jet $p_T > 500$ GeV, lepton and photon veto, $\Delta\phi(\text{jets}, \text{MET}) > 0.4$; eight MET bins from 1 to beyond 15 TeV.
- *Statistics*: binned profile likelihood with log-normal nuisances (process normalisations 15–20%, signal 30%, luminosity 2%, jet/MET scale 2%, MC statistics), auxiliary global observables, Asimov significance; toys reproduce the asymptotic median within 0.1σ.

3 Reach and uncertainty budget

Group	Variation	Δ discovery [TeV]	Δ exclusion [TeV]
Background	$\mu_R = \mu_F \times 2$ / $\times 0.5$ (smoothed)	−0.07 / −0.08	−0.07 / −0.09
	PDF CTEQ6L1 / NNPDF3.1	+0.09 / 0.00	+0.08 / 0.00
	$K \times 1.2$ / $\times 0.8$	−0.16 / +0.21	−0.21 / +0.26
Signal	$\sigma \times 0.7$ / $\times 1.3$	−0.48 / +0.34	−0.54 / +0.39
	luminosity NNPDF3.1 / mirrored	−0.66 / +0.71	−0.83 / +0.95
	greybody emission	−0.06	−0.19
Detector	response tail $\times 10$	−0.62	+0.37
	degraded pile-up suppression	−0.33	+0.28
	scale nuisance 5%	+0.03	−0.04
Total (quadrature)		+0.82 / −1.10	+1.16 / −1.03
Not in total: LL electroweak Sudakov estimate		+0.81	+0.91

Table 1: One-at-a-time shifts of the reach (nominal 33.4 / 35.0 TeV).

The reach is robust to background modelling because the signal falls tenfold per ~ 3 TeV of M_{ext}; the high-x parton luminosity dominates. Electroweak corrections (tens of percent at TeV p_T; Becher–Garcia i Tormo arXiv:1305.4202, Kühn et al. hep-ph/0507178) only lower the background and are not counted. Raw generator-variation samples give larger shifts (0 to -0.8 TeV) dominated by Monte Carlo fluctuations in the top MET bins: high-statistics tail samples are required for a real analysis.

4 Remnants and alternatives

Remnants are slow (β median 0.11, all below 0.5). Neutral remnants are pure missing energy (the default after colour discharge). Charged remnants would be highly ionising ($\sim 31\times$ MIP), arrive ~ 41 ns late at 1.5 m, and $\sim 40\%$ would stop in an assumed $1450\,\text{g}\,\text{cm}^{-2}$ of calorimeter (uncertain by a factor ~ 2 at low $\beta\gamma$). The exact likelihood-ratio separation at $S/B = 1$ requires 13 events against a GR hole with the same threshold, 560–670 against ADD graviton emission and 240–370 against dark-matter mediators, robust under the emission-model variations.

5 Current data

The 13 TeV ATLAS monojet likelihood excludes $M_{\text{ext}} < 3.5$–4.2 TeV (head-on trapping) or < 6.6–7.2 TeV (full trapping) for $n = 4$–6, $M_D = 2$–4 TeV; multi-object black-hole searches accept $\leq 0.4\%$ of model-U events and do not constrain it. A run on CMS Open Data is prepared but could not be executed here (network access to the open-data portal was not available).

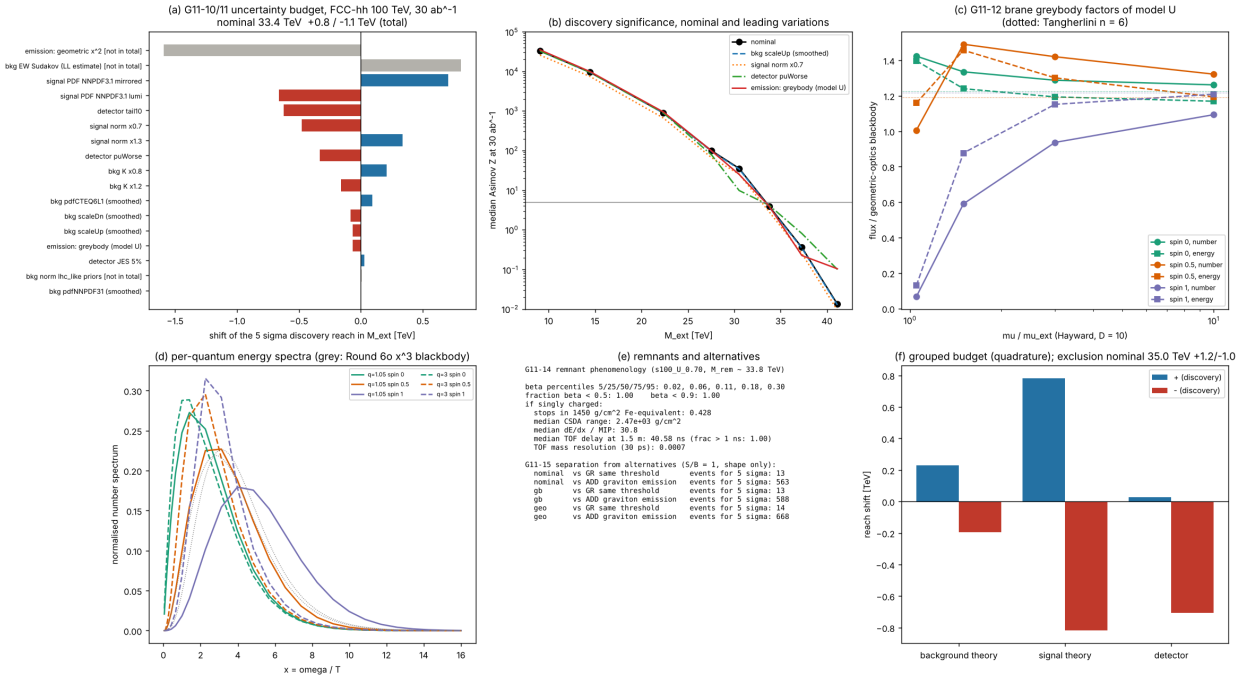


Figure 1: (a) Budget; (b) significance versus M_{ext} with leading variations; (c,d) greybody factors and spectra; (e) remnants and alternatives; (f) grouped budget.

Reproducibility

Configuration: `ROUND11/collider/analysis_config.json`; software versions, PDF hashes, process cards, seeds and sample hashes: `DATA/round11_manifest.json`; scripts `round10a.py`, `round11a.py`, `round11_greybody.py`, `round11b.py`.

References

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